Programming Concepts in Mechanical Engineering

Viva Questions and Answers (30 questions)

1. What is an algorithm?

Ans: A step-by-step list of instructions to solve a problem. It starts with Start and ends with Stop.

2. What is a flowchart? Name the main symbols.

Ans: A flowchart is a picture of an algorithm. Rounded box is Start/Stop, parallelogram is Input/Output, rectangle is Process and diamond is Decision.

3. What is the formula to convert Fahrenheit to Celsius?

Ans: C = (F - 32) x 5 / 9.

4. What is the formula for efficiency of a simple machine?

Ans: Efficiency = (Work output / Work input) x 100, where output = load x load distance and input = effort x effort distance.

5. What does input() return?

Ans: Always a string (str), even if you type a number.

6. What is type casting?

Ans: Changing a value from one data type to another, using int(), float(), str() or bool(). Example: int('5').

7. What is the difference between int(3.99) and round(3.99)?

Ans: int(3.99) cuts off the decimal and gives 3. round(3.99) goes to the nearest whole number and gives 4.

8. What is the difference between /, // and %?

Ans: / gives the decimal answer (7/2 = 3.5). // gives the whole-number answer (7//2 = 3). % gives the remainder (7%2 = 1).

9. What is a ValueError? How do you handle it?

Ans: It happens when a value has the right type but a wrong content, like int('abc'). Handle it with try and except so the program does not crash.

10. How do you swap two variables?

Ans: Use a temporary variable: temp = a, then a = b, then b = temp.

11. What is a while loop?

Ans: A loop that repeats its block again and again as long as its condition is true.

12. What is a factorial? What is 0!?

Ans: n! is the product of all whole numbers from 1 to n. 5! = 120. By definition 0! = 1. It is not defined for negative numbers.

13. What is a function? What does return do?

Ans: A function is a reusable block of code with a name. return sends a value back to the place where the function was called.

14. What is GCD? Which method did you use?

Ans: GCD is the largest number that divides both numbers exactly. We used Euclid's method: GCD(a, b) = GCD(b, a mod b). The GCD of 17 and 5 is 1.

15. How do you add, remove and sort items in a list?

Ans: append() adds, remove() deletes an item by value, and sort() arranges the list in order.

16. What is a CSV file?

Ans: A plain-text table where the values in a row are separated by commas. Python reads it with the csv module.

17. Why do we use with open(...)?

Ans: It closes the file automatically, even if an error happens.

18. What is the difference between reader and DictReader?

Ans: reader gives each row as a list. DictReader gives each row as a dictionary using the header names.

19. What is a class? What is an object?

Ans: A class is a blueprint. An object is a real item made from that class. Example: BankAccount is a class and your account is an object.

20. What are __init__ and self?

Ans: __init__ runs automatically when an object is created and sets its starting values. self means the current object.

21. How does the bank account program stop a wrong withdrawal?

Ans: It checks if the amount is more than the balance. If yes, it shows an insufficient balance message and does not reduce the balance.

22. Why is a motor driver needed with Raspberry Pi?

Ans: The GPIO pins give very little current and cannot run a motor directly. The driver (L298N) supplies the needed current.

23. What is an H-bridge?

Ans: A circuit of four switches that lets current flow through a motor in either direction, so the motor can turn forward or backward.

24. What is the use of the ENA pin and PWM?

Ans: ENA enables the motor. PWM (Pulse Width Modulation) switches it on and off quickly to control the motor speed.

25. What are the three main parts of a robot? How does an ultrasonic sensor measure distance?

Ans: Sensors, controller and actuators. The sensor sends a sound pulse and measures the echo time: distance = speed x time / 2.

26. Why is a voltage divider used on the ECHO pin?

Ans: ECHO gives 5 V, but Raspberry Pi GPIO pins accept only 3.3 V. The divider reduces the voltage safely.

27. What are mean, variance and standard deviation?

Ans: Mean is the average. Variance is the average of the squared differences from the mean. Standard deviation is the square root of the variance and shows how spread out the data is.

28. What are X-bar and R charts?

Ans: X-bar chart plots the average of each sample to watch the process centre. R chart plots the range (largest minus smallest) of each sample to watch the spread.

29. What are control limits (UCL and LCL)? Name some quality control tools.

Ans: Upper and Lower Control Limits are lines (usually mean plus or minus 3 standard deviations) that show whether a process is under control. Tools: histogram, Pareto chart, scatter plot and control chart.

30. What is stress? Why is stress higher in the narrow part of a plate?

Ans: Stress is force per unit area: sigma = P / A (unit N/mm2 or MPa). In the narrow part the area is smaller for the same load, so the stress is higher. A sudden change of section or a hole also causes stress concentration.